

WORK PROCEDURE

PRO-SEC-001A

PROCEDURE FOR BREAKTHROUGH HOLES

OBJECTIVES

The purpose of this procedure is to define safe methods for breakthrough holes underground.

APPLICATION

This procedure applies to all employees of MACHINES ROGER INTERNATIONAL.

When this procedure is followed, it must be bagged and hung on the barricade at the breakthrough location. The attached form must be signed at the beginning of each shift by the driller, helper or designated person who comes to inspect the site. If it is not the driller or helper who comes to do the inspection, the designated person must contact the driller to confirm that the procedure has been followed and signed.

INTERPRETATION

The staff with authority for the interpretation of this document is the Health and Safety superintendant. Any request for modification, revision or update must be made to this person.

RESPONSIBILITIES

The worker in charge of the drilling must make sure to apply this procedure and to respect the instructions that come from the engineering department.

PROCEDURE FOR INITIAL BREAKTHROUGH

A drill hole of any diameter or length, whether by diamond drilling, ITH drilling or other, must be identified on an engineering plan. Any excavation that may be crossed by this hole must also be indicated on the drilling plan.

When a hole is planned to breakthrough or possibly breakthrough, any excavation, the operator of the drill has two alternatives, A or B. He must apply his choice at least 5 meters (15 feet) before breaking through:

- (a) A minimum area of 15 meters (45 feet) around the estimated point of the hole must be closed by a barricade (fence, wire, orange barricade, etc.) and indicated by a "Dangerous Opening" sign and an elastic red ribbon "Drilling in progress" accompanied by a card indicating the risk.

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- b) A person must be sent to watch over the breakthrough area and warn workers working in this area. This person must receive instructions from the driller and agree on the time and location of the breakthrough. He should never be in the breakthrough area

N.B : In both cases, it will be necessary to ensure that all accesses to the area are barricaded or guarded by a worker

If there is any danger at the breakthrough site, (ex: near electrical cables, a powder magazine, a power station, etc.), the operator must immediately:

- Contact the responsible supervisor to take extra precautions.
- After the breakthrough, wait until the rods are completely out of the hole to release the 15-meter (45-feet) safety zone.
- If the hole breakthrough in a unattended gallery, a 5-meter (15-foot) perimeter around the hole should be delimited with an orange barrier and a "Dangerous Opening" sign.
- If the hole breakthrough in a traffic drift that can not be barricaded, the damaged area must be rehabilitated without delay and the arrival of the hole must be blocked with a wood plug of appropriate size, retained by a wire mesh attached to the roof or steel plate fixed in the wire mesh.
- A steel plate can also be installed according to the specifications initially established by the engineering team in anticipation of the casing of this hole.
- A red cone should be placed on the collar of the drilled hole, to avoid falling objects inside of the hole:

BORING

When boring a hole after initial drilling, either upward or downward, the following rules must be followed:

- If the hole breakthrough into a gallery, ramp or other, a deflector must be installed to deflect drilling muds on the edge of the wall.
- If drilling cutting can not be diverted, a perimeter of five (5) meters (15 feet) around the hole must be delimited 5 meters (15 feet) before the end of the hole to be reamed and the alternative must be done A or B, mentioned above.
- If the hole breakthrough in an non-attended drift or stope, the area must be closed by a barrier with the appropriate sign

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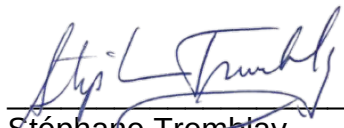
BREAKTHROUGH IN A STOPE

- The driller may not breakthrough or clean a hole, unless he has previously installed an orange barrier with a "dangerous opening" sign and his red ribbon "drill-in-progress" at the entrance of the stope.
- A "drilling in progress" sign must be installed at the entrance of the stop drawpoint.
- The two supervisors responsables (drilling and mucking) must inform all workers concerned that there is drilling and simultaneous mucking in the stope.
- They must check if the planned drilling they want to do could damage the scoop. Example: collapse, rock falling or dropped drill pipes.
- If there is an unexpected risk, an on-site full assessment should be done by the supervisors to determine whether it is appropriate to continue drilling and mucking at the same time.

NOTE :

- 1- It is strictly forbidden to work at the lower level of the stope when drilling is carried out at the upper level of the stope, except for a remote-controlled scoop.
- 2- When using the blockholer in the stope lower level, all drilling operations must be suspended.
- 3- If a hole breaks through when it is not supposed to, drilling must be stopped. The driller must immediately call his supervisor and access to the breakthrough hole must be closed. The mine will determine what further action is required.

Adds or modifications	Date	By
Add procedure signature in APPLICATION section Signature form added as appendix	December 21st 2023	S. Tremblay
Translation revision	April 2024	S. Tremblay C. St-Amour
Add not supposed breakthrough hole	July 23 rd 2024	S. Tremblay


 Stéphane Tremblay
 Health and Safety Superintendent.

PROCEDURE FOR BREAKTHROUGH HOLES

SIGNATURE FORM BEGINNING OF EACH SHIFT

DRILL LOCATION: _____

DRILL NUMBER: _____

BREAKTHROUGH LOCATION: _____

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (script) : _____

Signature of the person who verified: _____

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (sript) : _____

Signature of the person who verified : _____

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (sript) : _____

Signature of the person who verified : _____

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (sript) : _____

Signature of the person who verified : _____

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (sript) : _____

Signature of the person who verified : _____

PROCEDURE FOR BREAKTHROUGH HOLES

SIGNATURE FORM BEGINNING OF EACH SHIFT (PART 2)

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (script) : _____

Signature of the person who verified: _____

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (sript) : _____

Signature of the person who verified : _____

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (sript) : _____

Signature of the person who verified : _____

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (sript) : _____

Signature of the person who verified : _____

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (sript) : _____

Signature of the person who verified : _____

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (sript) : _____

Signature of the person who verified : _____

PROCEDURE FOR BREAKTHROUGH HOLES

SIGNATURE FORM BEGINNING OF EACH SHIFT (PART 3)

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (script) : _____

Signature of the person who verified: _____

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (sript) : _____

Signature of the person who verified : _____

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (sript) : _____

Signature of the person who verified : _____

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (sript) : _____

Signature of the person who verified : _____

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (sript) : _____

Signature of the person who verified : _____

Date : _____ Hour : _____ Day _____ Night _____

Name of the person who verified (sript) : _____

Signature of the person who verified : _____